Jannatul Ferdous

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Research Interests

Quantum optics and photonics; quantum sensing; quantum imaging; quantum metrology; multiphoton quantum optics; plasmonics and nanophotonics; photon-number-resolving detection; photonic waveguides; quantum information processing.

Education

Ph.D. in Physics (Quantum Photonics) Louisiana State University, Baton Rouge, LA
Expected 2027

M.S. in Physics University of Texas Rio Grande Valley, Edinburg, TX

M.S. in Physics University of Dhaka, Dhaka, Bangladesh

B.S. in Physics University of Dhaka, Dhaka, Bangladesh

Research Experience

Graduate Research Assistant — Ph.D. Research 2023–Present
Louisiana State University, Quantum Photonics Group

Project 1 — Quantum Statistical Plasmonic Metacrystals

- Designed and experimentally investigated room-temperature plasmonic metacrystals to engineer statistical bands and control the quantum statistics of multiparticle light.
- Used ANSYS Lumerical FDTD for electromagnetic modeling and design, and implemented photon-number-resolving measurements, quantum-state characterization, and confocal microscopy.

Project 2 — Quantum-Plasmonic Hydrogen Sensing

- Led the experimental development and system integration of a multiparticle quantum-plasmonic hydrogen sensing platform.
- Developed hydrogen calibration procedures and implemented LabVIEW-based synchronized data acquisition, photon-number-resolving detection, and optical characterization.
- Applied Fisher information analysis to quantify sensing performance, achieving $\sim$100$\times$ faster response and $\sim$46$\times$ improved detection limit compared with the classical protocol.

Project 3 — Quantum Reservoir Computing on a Plasmonic Platform

- Leading an ongoing experimental development of a plasmonic quantum reservoir for multidimensional information processing using OAM and polarization degrees of freedom.
- Implementing SLM-based spatial-mode manipulation, polarization control, PNR detection with APDs, time-tagged photon measurements, and optical correlation analysis.

Project 4 — Evolution of Multiphoton Quantum Coherence

- Conducted an experimental investigation of multiphoton quantum coherence during free-space propagation.
- Performed quantum-state tomography of the polarization degree of freedom, using experimental polarization measurements to reconstruct the density matrix with Python-based tomography methods.
- Applied SLM-based polarization control and photon-number-resolving detection, with theoretical analysis using MATLAB and Mathematica.

Project 5 — Multiphoton Quantum Imaging Using Natural Light

- Contributed to experimental development and data collection for a multiphoton quantum imaging platform using thermal light and photon-number-resolving detection.
- Performed raster scanning and experimental measurements for high-contrast image reconstruction under strong background noise using compressive sensing.

Additional Research Projects

Contributed to additional projects in plasmonic biosensing, quantum phase measurement using homodyne interferometry, photonic-waveguide information processing, and pseudo-thermal light-source development, applying optical system design, detector instrumentation, and quantum-state characterization. Additional experience includes DMD-based spatial mode control.

Graduate Research Assistant — M.S. Research 2019–2020
University of Texas Rio Grande Valley

Project 1 — Tunable Non-Hermitian Acoustic Filter

- Designed, fabricated, and experimentally characterized a tunable non-Hermitian acoustic filter.
- Developed the acoustic waveguide using additive manufacturing, performed acoustic measurements, and validated experimental behavior against COMSOL simulations.

Project 2 — Robust Zero-Energy Extended States

- Conducted theoretical and experimental investigations of robust zero-energy bulk states in a non-Hermitian acoustic system.
- Performed COMSOL modeling, acoustic waveguide characterization, controlled-disorder experiments, frequency-domain measurements, and transmission-spectrum analysis.

Selected Publications

First- and Co-First-Author Peer-Reviewed Articles

[1] Chenglong You, Riley B. Dawkins, **Jannatul Ferdous**, Mohammed Mehedi Hasan, Aadi Singh, Ziang Zhuang, Addison Wilberg, Ian Baum, Benjamin Bertoni, Mingyuan Hong, and Omar S. Magaña-Loaiza. "Quantum Statistical Plasmonic Metacrystals." ***Nature***, 655, 885–891 (2026). *Co-first Author.*

- Highlighted in *Nature News & Views.*
- Featured by the LSU College of Science and subsequently featured by *Phys.org.*
- Featured in *ScienceDaily* in a story highlighting the research.

[2] **Jannatul Ferdous**, Riley B. Dawkins, Cameron Bachar, Mingyuan Hong, Yang Mu, Khadijatu N. Alhassan, Chenglong You, Kevin M. McPeak, and Omar S. Magaña-Loaiza. "Scalable Quantum Sensing with Robust Multiparticle Plasmonic Systems." ***Nature Communications***, *Accepted in principle*, 2026. *First Author.*

[3] **Jannatul Ferdous**, Mingyuan Hong, Riley B. Dawkins, Fatemeh Mostafavi, Alina Oktyabrskaya, Chenglong You, Roberto de J. León-Montiel, and Omar S. Magaña-Loaiza. "Emergence of Multiphoton Quantum Coherence by Light Propagation." ***ACS Photonics***, 11, 3197–3204 (2024). *First Author.*

[4] **Jannatul Ferdous**, Cem Yuce, Andrea Alù, and Hamidreza Ramezani. "Observation of Robust Zero-Energy Extended States." ***Physical Review Applied***, 19, L061002 (2023). *First Author.*

Co-Authored Peer-Reviewed Articles

[5] S. Puri, **J. Ferdous**, A. Shakeri, A. Basiri, M. Dubois, and H. Ramezani. "Tunable Non-Hermitian Acoustic Filter." ***Physical Review Applied***, 16, 014012 (2021). *Co-first Author.*

[6] Mingyuan Hong, Mario A. Quiroz-Juarez, Riley B. Dawkins, Fatemeh Mostafavi, **Jannatul Ferdous**, Addison Wilberg, Blas M. Rodriguez-Lara, Justin Woodring, Chenglong You, Roberto de J. León-Montiel, and Omar S. Magaña-Loaiza. "Multiphoton Quantum Reservoirs For Robust Multidimensional Computing." ***Advanced Science***, e77225 (2026).

[7] Fatemeh Mostafavi, Mingyuan Hong, Riley B. Dawkins, **Jannatul Ferdous**, Ian Baum, Rui-Bo Jin, Roberto de J. León-Montiel, Chenglong You, and Omar S. Magaña-Loaiza. "Multiphoton Quantum Imaging Using Natural Light." ***Applied Physics Reviews***, 12, 011424 (2025).

[8] Chenglong You, Mingyuan Hong, Fatemeh Mostafavi, **Jannatul Ferdous**, Roberto de J. León-Montiel, Riley B. Dawkins, and Omar S. Magaña-Loaiza. "Isolating the Classical and Quantum Coherence of a Multiphoton System." ***PhotoniX***, 5, 39 (2024).

Conference Presentations

Plasmonic Metacrystals for Quantum Control — FiO+LS 2026, Rochester, NY, September 27–October 1, 2026, Oral Presentation; *Conference Paper to be Published*

Robust Multiparticle Quantum Plasmonics for Gas Sensing — FiO+LS 2026, Rochester, NY, September 27–October 1, 2026, Poster Presentation; *Conference Paper to be Published*

Nonclassical Multiphoton Plasmonic Sensing — IEEE RAPID (2025), Oral Presentation

Observation of the van Cittert-Zernike Modification of Quantum Coherence — LAOP (2024), Oral Presentation

Quantum Photonic Technologies Based on Natural Light — LSU Institute for Energy Innovation (2024), Oral Presentation

Inducing Quantum Coherence by Light Propagation — LSU Graduate Research Conference (2023), Oral Presentation

Teaching Experience

Graduate Teaching Assistant
Louisiana State University, LA, US 2021–2023
University of Texas Rio Grande Valley, TX, US 2020–2021

- Instructed undergraduate laboratories in experimental techniques, instrumentation, data analysis, and simulation-based experiments.
- Delivered lectures, coursework, and online laboratory instruction.

High School Physics Teacher
Cardiff International School, Dhaka, Bangladesh 2018–2019

- Taught theoretical and experimental Physics to O-Level and A-Level students under the British curriculum.

Technical Skills

Quantum Optical Techniques	Photon-number-resolving detection, optical characterization of materials, quantum-state characterization, quantum-state tomography, quantum sensing, quantum imaging, quantum metrology
Optical Systems	Interferometry, advanced optical system design, free-space optics, confocal microscopy, polarization control, OAM manipulation
Detectors & Instrumentation	SNSPD, APD, SLM, DMD, lasers, motorized stages, optical components, data acquisition systems, time tagger
Photonics	Plasmonics, nanophotonics, photonic waveguides, LSPR/SPR
Computational Modeling	COMSOL Multiphysics, ANSYS Lumerical FDTD
Programming & Automation	Python, MATLAB, Mathematica, LabVIEW, LaTeX
Design & Fabrication	Blender 3D modeling, 3D printing, experimental hardware prototyping, FIB fabrication

Professional Service

Journal Referee: Physical Review A; Quantum Science and Technology; Journal of the Optical Society of America B; Europhysics Letters.

Leadership & Service

International Student Liaison 2022–2023
Department of Physics, Louisiana State University

- Represented international students, coordinated communication with faculty and university leadership, and initiated efforts to gather student feedback on departmental and university-wide concerns.

Media Coordinator 2025–2026
Bangladesh Students Association, Louisiana State University

- Managed cultural event and social media coverage, creating event flyers and digital content.

Awards & Honors

Presidential Graduate Research Assistantship Award 2019
University of Texas Rio Grande Valley

References

Dr. Omar S. Magaña-Loaiza
Associate Professor
Louisiana State University
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Dr. Kevin M. McPeak
Associate Professor of Chemical Engineering
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